MONOGRAPH OF THE ZENODO SERIES

Package Formula and the Electroweak Core

Hexagonal Structure, the Słowik Phenomenon and a Periodic Table of Black Holes on the Macro Scale

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Monograph of the Zenodo series — merged edition
Source papers dated 11.07.2026 – 06.08.2026 (Table 1).

Abstract

This work presents an original, coherent gravitational concept called the Package Formula. The fundamental assumption is the existence of a minimal, stable structural unit of collapse: 1 Słowik Package (1 P) $= 2.8\,M_{\odot} = 8.27$ km of Schwarzschild radius. On this basis we build: (I) the mathematical Package Formula $R_s = N \times 8.27$ km, (II) the dynamic Słowik Phenomenon as a mechanism rejecting excess mass into the accretion disk, (III) a geometric hexagonal model and a macroscopic periodic table ($n = 2$, $n = 6$), (IV) a microscopic electroweak stabilization mechanism removing the singularity ($R_{\text{core}} \approx 15.8$ m), and (V) a unification of the continuous and quantum growth regimes with the role of the fundamental Higgs constants and the W/Z bosons.

Source papers (Zenodo series)

Table 1: Papers merged in this monograph; numbering follows the series listing.

No.	Title	Date	DOI
1	Mechanism for Halting Gravitational Collapse via Reverse Higgs Phase Transition	11.07.2026	doi:10.5281/zenodo.21314918
2	Microscopic Quantum Mechanisms of Stabilization / *Mikroskopowe kwantowe mechanizmy stabilizacji*	27.07.2026	doi:10.5281/zenodo.21624309
3	Package Formula for Schwarzschild Radius	27.07.2026	doi:10.5281/zenodo.21625028
4	Package Formula: Hexagonal Structure and Periodic Table of Black Holes	27.07.2026	doi:10.5281/zenodo.21625386
5	Hypothesis of Dynamic Synthesis and Rejection of Excess Mass (Słowik Phenomenon)	27.07.2026	doi:10.5281/zenodo.21625199
6	Calculation of Schwarzschild Radius and Package Method	28.07.2026	doi:10.5281/zenodo.21651625
7	Microscopic Quantum Mechanisms of Stabilization of Relativistic Electroweak Objects $2.8\,M_{\odot}$	06.08.2026	doi:10.5281/zenodo.21827082

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1 The Package Formula for the Schwarzschild radius

Source papers [3], [6].

1.1 Handwritten source note

TOV (Tolman–Oppenheimer–Volkoff)
MINIMUM MASS TO FORM A BLACK HOLE: 2.5–2.6 SOLAR MASSES GIVES 2.8
SCHWARZSCHILD FORMULA: $R_s = 2GM/c^2$
FOR $2.8\,M_\odot \to 8.27$ km
PACKAGE FORMULA (MARCIN SŁOWIK): $N \to$ (1 PACKAGE = 2.8 solar masses),
$R_s = N \times 8.27$

The Tolman–Oppenheimer–Volkoff limit [9, 10] and the Schwarzschild formula [8] are the starting points.

1.2 Validation

$$\boxed{R_s(N) = N \times 8.27\ [\mathrm{km}]}, \qquad N = M/2.8\,M_\odot. \tag{1.1}$$

Table 1.1: Comparison of the classical and the Package formula [25, 26].

Object	**Mass** ($M_\odot$)	N	**Classical** R_s	**Package** R_s
Minimum black hole	2.8	1.0	8.27 km	8.27 km
Cygnus X-1	21.2	7.57	62.61 km	62.61 km
Sagittarius A*	4 300 000	1 535 714	12.7 mln km	12.7 mln km

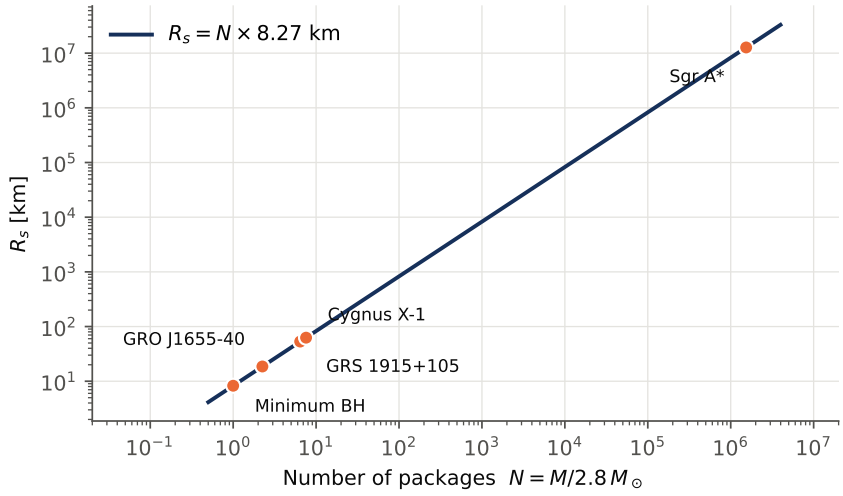


Figure 1.1: The Package Formula on a log–log scale: the objects of Tables 1.1 and 3.1 lie on a single straight line.

2 The Słowik Phenomenon — dynamic synthesis and rejection of excess

Source paper [5].

1. **Threshold quantization:** $1\,\mathrm{P} = 2.8\,M_\odot$ is an indivisible portion; there are no intermediate states (e.g. a 0.5 P black hole).
2. **Tendency toward the energy minimum.**
3. **Violent rejection of the excess** by the collapse shock wave.
4. **Buffer formation** (the accretion disk).
5. **Evolution and smooth growth** $N > 1$ by accretion from the buffer.

Analogy with the mass defect in nuclear fusion [21]: the excess mass is not absorbed but rejected. The accretion disk is a physically required safety valve.

$$\boxed{M_{\mathrm{total}} = n \times (2.8\,M_\odot) + M_{\mathrm{disk}}} \tag{2.1}$$

Table 2.1: Analogy to nuclear fusion and mass defect.

Feature	**Nuclear fusion (helium synthesis)**	**Słowik Phenomenon (gravitational collapse)**
System tendency	Achieving the lowest, most stable energy state of the atomic nucleus.	Achieving the most stable, compact gravitational state ($1\,\mathrm{P} = 2.8\,M_\odot$).
Reaction to excess	Excess mass cannot remain in the new nucleus — released as energy ($E = mc^2$).	Excess mass of the star is not immediately absorbed — it is ejected by the collapse explosion force.
Final effect	A stable quantized α particle (helium nucleus) is formed and radiation emitted.	A stable black-hole core ($1\,\mathrm{P}$) is formed, and a buffer (accretion disk) forms around it.

The Słowik Phenomenon hypothesis sheds new light on why newly formed stellar black holes show such high consistency in their initial masses. It shows that the accretion disk is not merely a random element of the environment, but a physically required safety buffer that receives excess matter ejected during the quantized phase transition that black-hole births are.

3 Hexagonal structure and periodic table

Source paper [4].

Table 3.1: Macroscopic periodic table [27, 28, 29, 30, 31, 32].

Object	**Mass** ($M_\odot$)	n	**Configuration / spin** a_*	**Analogue**
Base unit	2.8	1	quantum, —	Hydrogen ^{1}H
GRO J1655-40	∼6.3	2	binary, 0.65–0.80	Helium ^{4}He
GRS 1915+105	10–18	6	hexagon, 0.98–1.00	Carbon ^{12}C

GRS 1915+105: $18/2.8 \approx 6$, so $6 \times 2.8 = 16.8\,M_\odot$. The number 6 denotes hexagonal symmetry as in the carbon-^{12}C nucleus; spin $a_* \to 1.0$, 1150 rev/s.

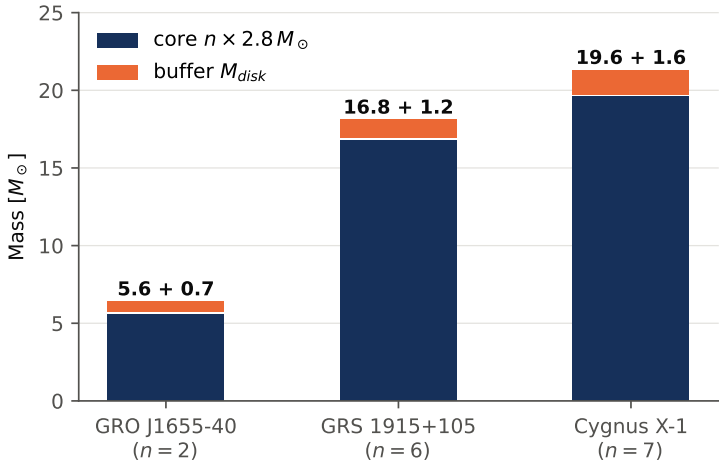


Figure 3.1: Decomposition of mass into the core $n \times 2.8\,M_\odot$ and the buffer M_{disk} (for GRS 1915+105 the upper estimate $18\,M_\odot$ is used; Cygnus X-1: $19.6 + 1.6$).

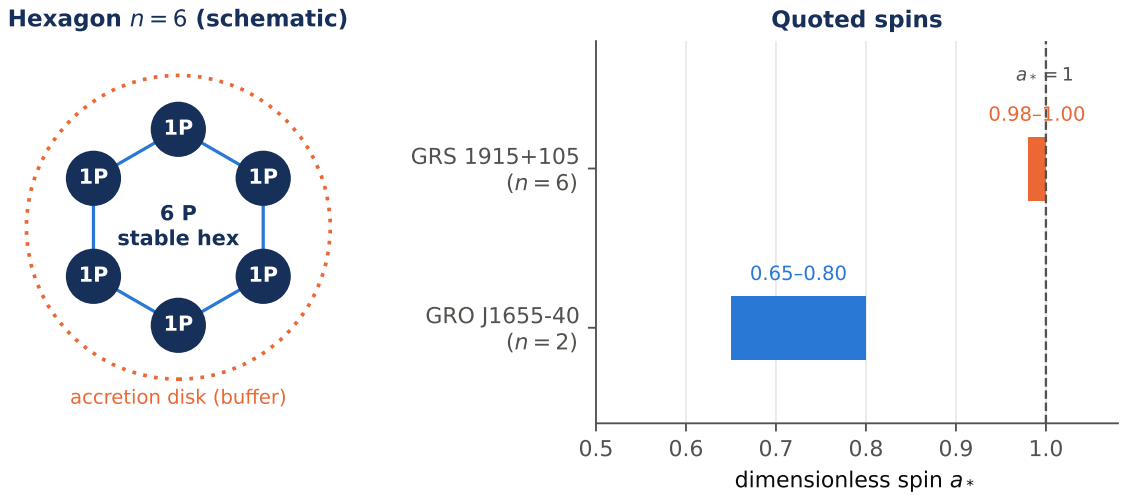


Figure 3.2: Left: schematic of the $n = 6$ hexagon. Right: measured spins of the objects in Table 3.1.

4 Microscopic mechanisms — the 15.8 m core

Source papers [1], [2], [7].

4.1 Setting

Extreme energy density and temperature during gravitational collapse trigger a reverse Higgs phase transition (symmetry restored to $v = 0$). The resulting sudden loss of particle rest mass and the emergence of massless relativistic radiation pressure halt further collapse and prevent singularity formation.

Above the TOV limit (about 2.2–2.6 $M_\odot$) neutron-degeneracy pressure can no longer balance gravity. Instead of a point singularity [11, 12], the model proposes a structured core of radius 15–48 m with four layers tied to Standard-Model energy scales. For $M = 2.8\,M_\odot$:

$$R_{\text{core}} = \frac{G\,M\,m_0}{E_H}, \tag{4.1}$$

$$\rho_{\text{core}} = \frac{M}{R_{\text{core}}^3}, \tag{4.2}$$

$$p_{\text{core}} = \frac{1}{3}\,\rho_{\text{core}}\,c^2 = \frac{1}{3}\,\frac{M}{R_{\text{core}}^3}\,c^2, \tag{4.3}$$

Parameter:

$$\eta = \frac{p_{\text{core}}\,R_{\text{core}}^4}{G\,M^2}. \tag{4.4}$$

Substituting:

$$\eta = \frac{\frac{1}{3}\frac{M}{R_{\text{core}}^3}\,c^2 \cdot R_{\text{core}}^4}{G\,M^2} = \frac{1}{3}\,\frac{M\,c^2\,R_{\text{core}}}{G\,M^2} = \frac{1}{3}\,\frac{c^2\,R_{\text{core}}}{G\,M}. \tag{4.5}$$

Further:

$$R_{\text{core}} = \frac{G\,M\,m_0}{E_H} \;\Rightarrow\; \eta = \frac{1}{3}\,\frac{c^2}{G\,M}\cdot\frac{G\,M\,m_0}{E_H} = \boxed{\frac{1}{3}\,\frac{c^2\,m_0}{E_H}} \tag{4.6}$$

so η does not depend on M or G. Numerically $\eta \approx 1.27 \times 10^{-3}$ for $m_0 = 1.675 \times 10^{-27}$ kg and $E_H = 246$ GeV.

4.2 Mechanisms

Inverted Higgs transition ($v = 0$) [15, 16, 17]: $W^\pm$ (80.4 GeV) and Z^0 (91.2 GeV) "melt" at 160–246 GeV, particles lose rest mass and the matter behaves as an ultra-relativistic gas, $P = \frac{1}{3}u$ (radiation pressure). **Beta processes and tunnelling** [19, 20]: $n \leftrightarrow p + e^- + \bar{\nu}_e$ (reverse: $p + e^- \to n + \nu_e$) produces a macroscopic density of positive charge, hence Coulomb repulsion as a second pressure vector. **Relativistic spin:** the Kerr solution [13] stretches the singular point into a ring; frame dragging [14] prevents formation of a standard horizon (stated as a conclusion of the model). Result: $R_{\text{core}} \approx 15.8$ m versus $R_s \approx 8.27$ km.

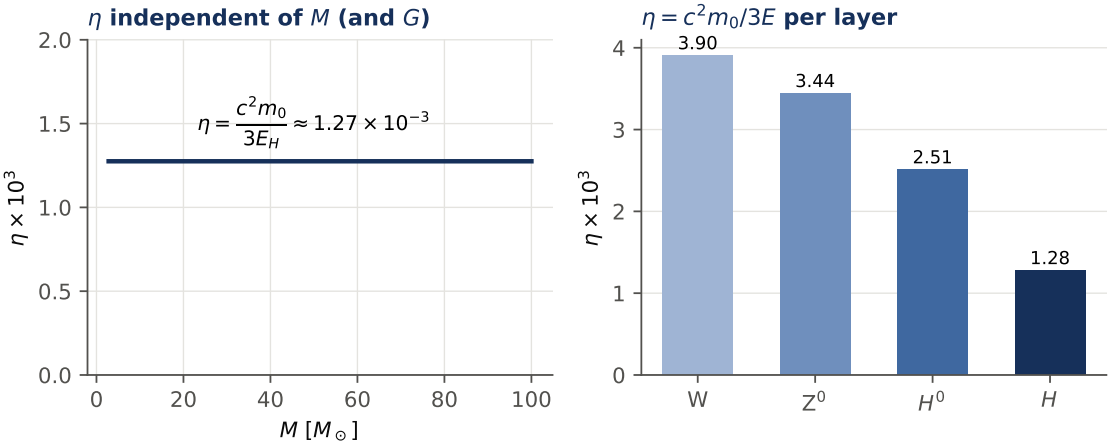


Figure 4.1: The parameter η: independent of M (left), and for each electroweak energy scale (right).

4.3 Four layers of the core

Constants used: $G = 6.674 \times 10^{-11}$, $M = 5.5692 \times 10^{30}$ kg, $m_0 = 1.675 \times 10^{-27}$ kg, $c = 3 \times 10^8$ m/s, $1\,\mathrm{eV} = 1.602 \times 10^{-19}$ J.

Table 4.1: Energy scales [18], radii, densities and radiation pressures of the four core layers ($M = 2.8\,M_\odot$).

Layer	**Scale**	R_{core} [m]	ρ [kg/m^3]	$u = \rho c^2$ [J/m^3]	p [Pa]
1	W 80.38 GeV	48.35	4.9272×10^{25}	4.4345×10^{42}	1.4782×10^{42}
1	Z^0 91.19 GeV	42.62	7.1937×10^{25}	6.4743×10^{42}	2.1581×10^{42}
2	H^0 125.1 GeV	31.06	1.8586×10^{26}	1.6727×10^{43}	5.5758×10^{42}
2	H 246 GeV	15.80	1.4120×10^{27}	1.2708×10^{44}	4.2359×10^{43}

For comparison, a neutron star [24] has radius 10–13 km and density 10^{17}–10^{18} kg/m^3. **Phase 1** (layers of W, Z^0): bosons of the weak interaction (beta decay, tunnelling) are melted by gravitational squeeze. **Phase 2** (layers of H^0, H): mass ceases to be a constant property; the 15 m region restores the state of the Universe about 10^{-12} s after the Big Bang.

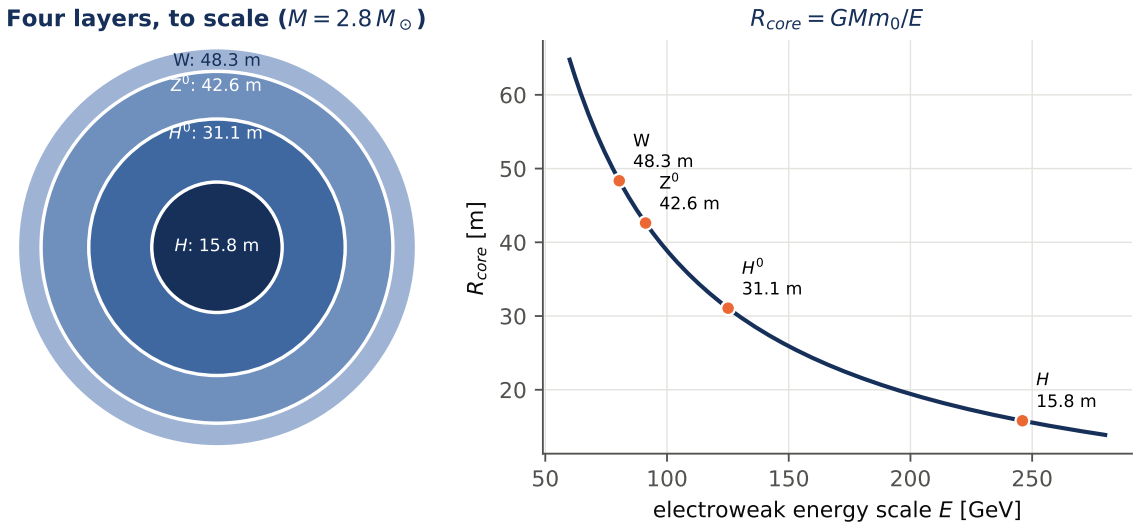


Figure 4.2: Left: the four layers drawn to scale. Right: core radius as a function of the electroweak energy scale.

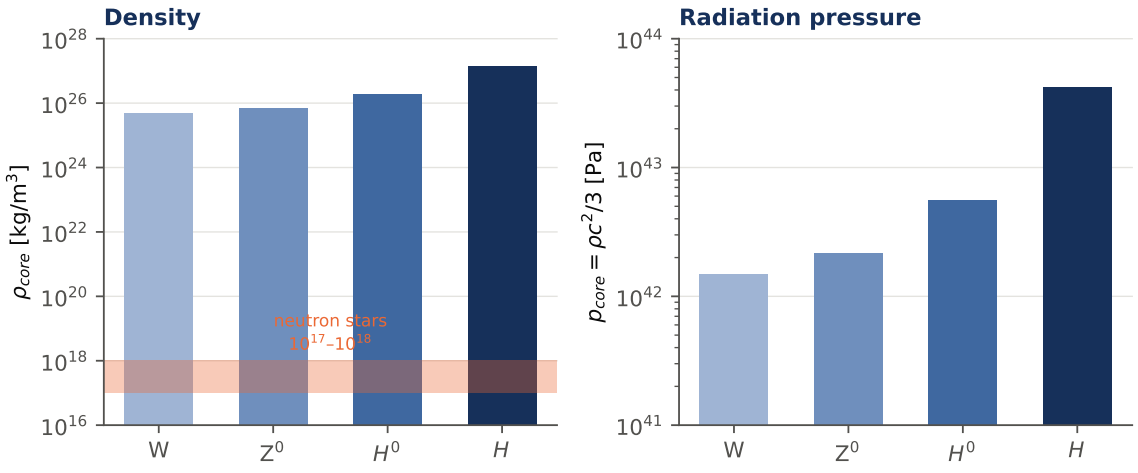


Figure 4.3: Density and radiation pressure of each layer (log scale); the shaded band marks neutron-star densities.

5 Unification of the continuous and quantum regimes

Two radii: $R_s(\text{base}) = 8.27$ km and $R_{\text{core}} = 15.8$ m. Two growth regimes: **Regime A – buffer** ($0 < M_{\text{disk}} < 2.8$) – continuous, 95% of objects, e.g. Cygnus X-1 $= 19.6$ $[n = 7]$ $+1.6$ buffer. **Regime B – quantum** ($M_{\text{disk}} \geq 2.8$) – discontinuous, barrier $E \approx 5.01 \times 10^{47}$ J $= Mc^2$, $n \to n + 1$, flare/jet.

As in nuclear physics the magic numbers $2, 8, 20, 28, 50, 82, 126$ [22, 23] mean increased stability, so in the macro scale $n = 2$ (GRO J1655-40) and $n = 6$ (GRS 1915+105) mean hexagonal configurations of increased rotational stability explaining extreme spin $a_* \to 1.0$.

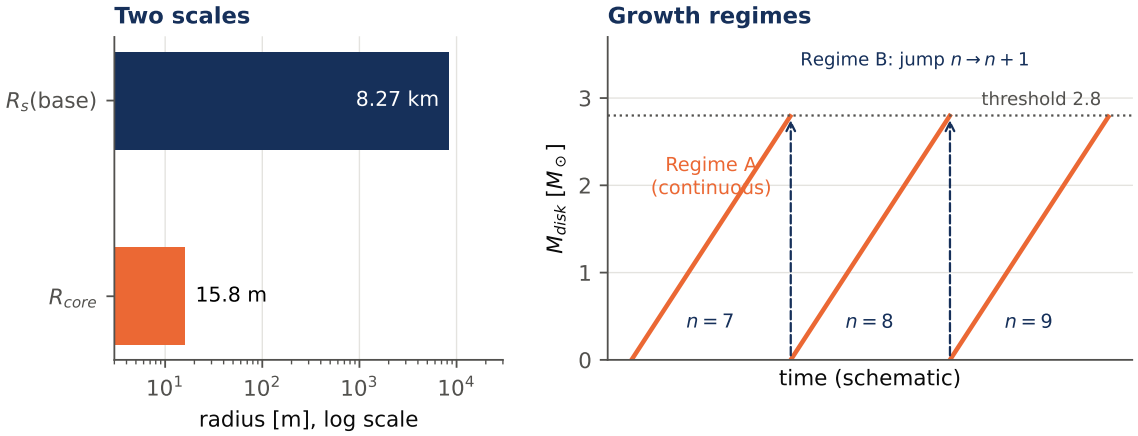


Figure 5.1: Left: the two radii of the model on a logarithmic scale. Right: schematic of regimes A and B: the buffer grows continuously up to $2.8\,M_\odot$, then a jump $n \to n + 1$ occurs.

Table 5.1: Model parameters.

Parameter	**Value**	**Role**
1 P threshold	$2.8\,M_\odot$	mass quantum
Base R_s	8.27 km	unit
R_{core}	15.8 m	no singularity
Higgs	246 GeV	$v = 0$ threshold
W/Z	80.4–91.2 GeV	pressure
Barrier	5.01×10^{47} J	jump $N \to N + 1$

Final conclusions

$R_s = N \times 8.27$ km – a didactic unit. The Słowik Phenomenon – $M_{\text{total}} = n \times 2.8 + M_{\text{disk}}$, the buffer as a safety valve. Hexagon $n = 2, 6$ – magic numbers. R_{core} 15.8 m and η independent of G – elimination of the singularity. Regimes A/B – full dynamics.

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